

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T31 · Coordinate Geometry

With Answers

Questions with full worked solutions

■ 25 pages

FAMILY MAP

Coordinate Geometry

F01	Determine coordinates from geometric information	6 Q 28 marks
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F02	Use the midpoint relationship	1 Q 2 marks
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F03	Read and plot Cartesian coordinates	3 Q 13 marks
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Question 28

4MA1-2H Q24

ORIGINAL QUESTION

6 marks

24 $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(..... ,)

Worked Solution 28

Coordinates of the fourth vertex of a kite

MODULAR UNIT 1**6 marks****Find AC**

$$m_{AC} = \frac{-5 - 10}{4 - (-2)} = -\frac{5}{2}, \quad y = 5 - \frac{5}{2}x.$$

Use the perpendicular diagonal

$$m_{BD} = \frac{2}{5}, \quad y - 4 = \frac{2}{5} \left(x + \frac{27}{5} \right).$$

Find the diagonal intersection

$$AC \cap BD = \left(-\frac{2}{5}, 6 \right).$$

Reflect B in AC

$$D = 2 \left(-\frac{2}{5}, 6 \right) - \left(-\frac{27}{5}, 4 \right) = \left(\frac{23}{5}, 8 \right).$$

Final answer

$$D = (4.6, 8)$$

Original question 28

4MA1-2H Q22 Q22

ORIGINAL QUESTION**7 marks**

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D .

(.....,)

(.....,)

Worked solution

Family: Determine coordinates from geometric information • Variant: Kite or symmetric vertex coordi-

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 28 MODULAR UNIT 1
Chapter: Coordinate Geometry 7 marks
Q22 – Chapter: Coordinate Geometry
Family: Determine coordinates from geometric information • Variant: Kite or symmetric vertex coordi-
nates
Method
The gradient of AM is $(2 - 4)/(0 - (-3)) = -2/3$.
Why: Use the two given coordinates.
Method
The kite diagonals are perpendicular, so BD has gradient $3/2$ and equation $y = (3/2)x + 2$.
Why: Use the negative reciprocal and point M.
Method
For B or D, $(x + 3)^2 + (y - 4)^2 = 6.5^2$.
Why: Each point is 6.5 units from A.
Method
Substitute $y = (3/2)x + 2$: $(x + 3)^2 + ((3/2)x - 2)^2 = 42.25$.
Why: Combine the line and distance conditions.
Method
Expanding gives $3.25x^2 + 13 = 42.25$.
Why: Expand and collect; the linear x terms cancel.
Method
Thus $x^2 = 9$, so $x = 3$ or $x = -3$; the corresponding y-values are 6.5 and -2.5.
Why: Solve the quadratic and substitute into the line.
Method
Since B has positive x-coordinate, B = (3, 6.5) and D = (-3, -2.5).
Why: Use the final labelling condition.
Final answer
B: B = (3, 6.5)
Final answer
D: D = (-3, -2.5)

FINAL ANSWER

- 1 The point A has coordinates $(5, -4)$
The point B has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of AB .

(.....,)
(2)

Worked solution

January 2020 | Question 1 | 2 marks

Family: Use the midpoint relationship • Variant: Midpoint from two endpoints

Method / working

1. Midpoint $x = (5+13)/2 = 9$.
 2. Midpoint $y = (-4+1)/2 = -1.5$.
- Final answer: (9, -1.5)

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

Worked solution

January 2020 | Question 22 | 6 marks

Family: Determine coordinates from geometric information | Variant: Coordinates from isosceles and gradient constraints

METHOD / WORKING

1. $M = ((j+6)/2, (15+k)/2)$.
2. Gradient $HM=2$ gives $(k+13)/(j+14)=2$, so $k=2j+15$.
3. $JK^2 = 80$ gives $(6-j)^2 + (k-15)^2 = 80$.
4. Substitute k : $5j^2 - 12j - 44 = 0$.
5. The condition $j < 0$ selects $j = -2$.
6. Then $k = 2(-2) + 15 = 11$.

FINAL ANSWER

$j = -2, k = 11$

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

Worked solution

January 2021 | Question 18 | 4 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Method / working

1. $AB=(12,5)$, so $b=14$ from the rectangle translation.
2. Gradient $AB=12/5$, hence gradient $BC=-5/12$.
3. Use $C=(19,a)$ to obtain $a=2.5$ and then $D=(14,-9.5)$.
4. State all three coordinates values.

Final answer: $a=2.5$, $b=14$, $c=-9.5$

22 $ABCD$ is a rhombus.

The diagonals, AC and BD , intersect at the point M .

The coordinates of M are $(6, -11)$

The points A and C both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through B and D intersects the y -axis.

(.....,)

(Total for Question 22 is 4 marks)

Answer reference | May-Nov 2020 | Question 22

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

22	$y = -\frac{7}{2}x(+10)$ or (gradient $\Rightarrow -\frac{7}{2}$		4	B1 for correct gradient which may be seen in an equation. Condone $-\frac{7}{2}x$
	$-\frac{7}{2}, m = -1$ or $(m \Rightarrow) \frac{2}{7}$			M1 fit their gradient for use of $m_1 \times m_2 = -1$
	$-11 = \frac{2}{7} \times 6 + c$ or $y - -11 = \frac{2}{7}(x - 6)$ oe			M1 fit dep on M1
		$\left(0, -\frac{89}{7}\right)$		A1 accept $\left(0, -12\frac{5}{7}\right)$ must be exact values
				Total 4 marks

Worked solution

May-Nov 2020 | Question 22 | 4 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Method

In a rhombus, the diagonals are perpendicular and bisect each other.

The diagonal AC lies on

$$2y+7x=20$$

$$2y=-7x+20$$

$$y=-\frac{7}{2}x+10$$

So the gradient of AC is

$$-\frac{7}{2}$$

Therefore the gradient of BD is

$$\frac{2}{7}$$

The line BD passes through

$$M=(6,-11)$$

So

$$y+11=\frac{2}{7}(x-6)$$

To find where this line intersects the y-axis, put $x=0$:

$$y+11=\frac{2}{7}(0-6)$$

$$y+11=-\frac{12}{7}$$

$$y=-11-\frac{12}{7}$$

$$y=-\frac{89}{7}$$

Answer: $\text{displaystyle } (0,-\frac{89}{7})$

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.

The point D has coordinates $(8, 7)$

The straight line L passes through the points B and D

The straight line L is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.



(Total for Question 23 is 6 marks)

Answer reference | January 2023 | Question 23

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

23	$y = \frac{3}{5}x \left(+ \frac{6}{5} \right)$ or $y = 0.6x(+1.2)$ or (gradient =) $\frac{3}{5}$ or 0.6 oe		6	M1 for correct gradient which may be seen in an equation. Condone $\frac{3}{5}x$ or $0.6x$
	$k = -2$			B1 for $k = -2$
	$\left(\frac{-2+8}{2}, \frac{1+7}{2} \right)$ oe or (3, 4)			M1 for finding the midpoint (use of their k where $k < 0$)
	$\frac{3}{5}m = -1$ or $(m =) -\frac{5}{3}$			M1ft their gradient for use of $m_1 \times m_2 = -1$ Allow $-\frac{5}{3} = -1.67$ or better
	$"4" = -\frac{5}{3}"x"3" + c$ or $c = 9$ or $y - "4" = -\frac{5}{3}"(x - "3")$			M1 dep on M3
	Working required	$5x + 3y = 27$		A1 allow equation in any form where p, q and r are integers
Total 6 marks				

Worked solution

January 2023 | Question 23 | 6 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Official final mark-scheme pages are embedded immediately after this question.

6 The points A and B are on a coordinate grid.

The coordinates of A are $(6, 4)$

The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 6 is 3 marks)

Answer reference | January 2023 | Question 6

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Question	Working	Answer	Mark	Notes
6	$[k =] \frac{6+17}{2}$ or $[k =] 6 + \frac{17-6}{2}$ oe or $[j =] 4 + 2(15-4)$ or $[j =] 15 + (15-4)$ or $\frac{4+j}{2} = 15$ oe		3	M1
	<i>Correct answers score full marks (unless from obvious incorrect working)</i> <i>1 correct answer will score M1A1 and both will score M1A1A1</i>	26		A1
		11.5		A1 oe eg $\frac{23}{2}$ both answers the wrong way round scores M1A1 unless the correct answers are clearly labelled in working space

Worked solution

January 2023 | Question 6 | 3 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Official final mark-scheme pages are embedded immediately after this question.

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

Worked solution

November 2025 | Question 20 | 3 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Method

Line L has equation

$$y=4x+7$$

So its gradient is

4

Line M is perpendicular to L, so its gradient is

$$-\frac{1}{4}$$

Line M passes through (8,1).

$$y-1=-\frac{1}{4}(x-8)$$

$$y-1=-\frac{1}{4}x+2$$

$$y=-\frac{1}{4}x+3$$

Answer:
$$y=-\frac{1}{4}x+3$$

- 1 The point P has coordinates $(3, 4)$
The point Q has coordinates $(9, 16)$

M is the midpoint of the line PQ

Find the coordinates of M

(.....,)

(Total for Question 1 is 2 marks)

Worked solution

November 2025 | Question 1 | 2 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Solution

$P=(-3,0)$, $Q=(9,16)$

The midpoint is

$((-3+9)/2, (0+16)/2)$

$= (3,8)$

Answer: $(3,8)$.